

MACHINE LEARNING FOR MATERIALS

MSE 404/MSE504

Instructor: Sergei V. Kalinin

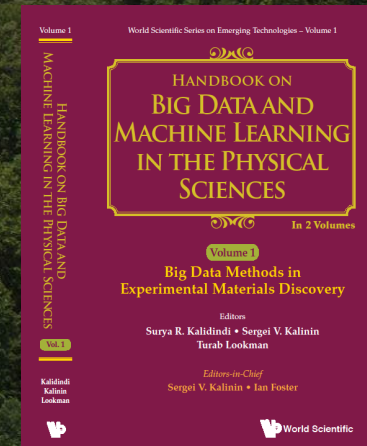
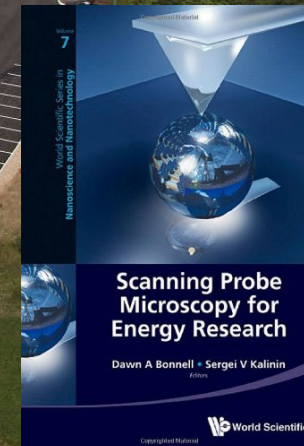
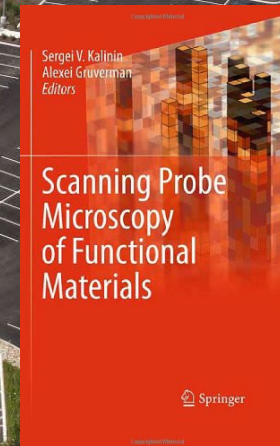
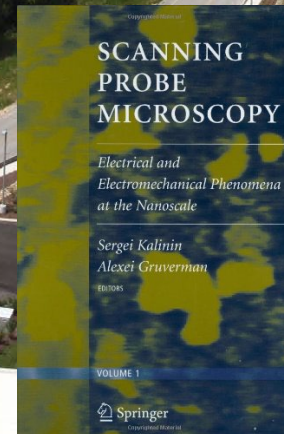
**Times and locations: 10:20 am - 11:10
am MWF, Ferris Hall 511**

2002 - 2022

Since 2022

2022 - 2023

amazon



Materials Science cannot change overnight....
... but what is happening now comes very close

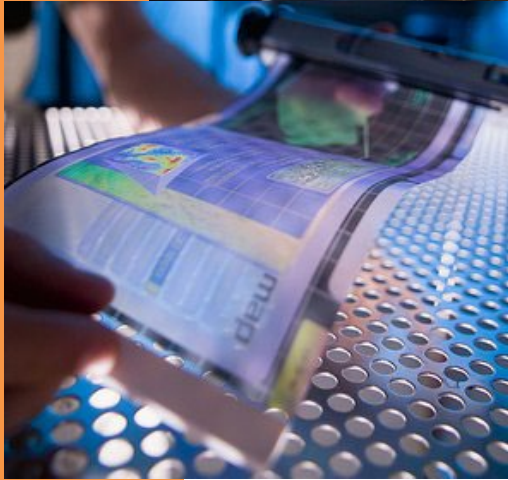
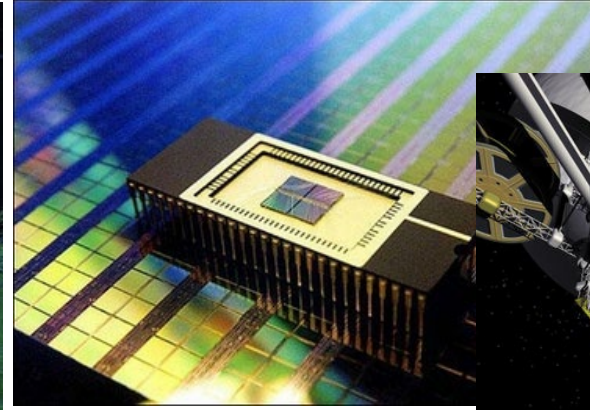
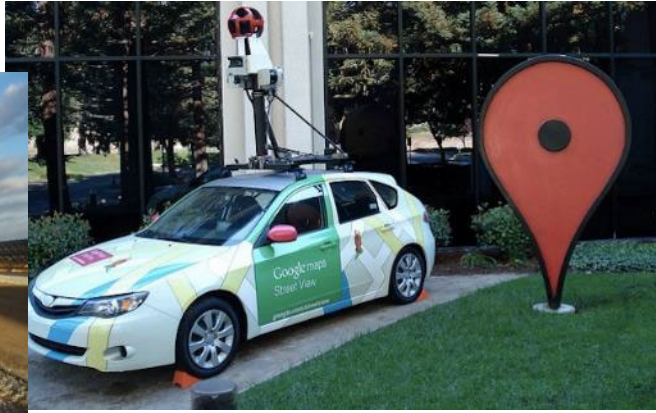


More than 1.5 \$B VC
investment over the
last year!

Prometheus: \$6B

- Lila Scientific
- Periodic Labs
- Radical AI
- Phase Tree
- Aionics
- Eionics
- Atinary
- Google
- Amazon
- ...

The World is Material Opportunity



Predicting crystal structure by merging data mining with quantum mechanics

CHRISTOPHER C. FISCHER¹, KEVIN J. TIBBETTS¹, DANE MORGAN² AND GERBRAND CEDER^{1*}

¹Department of Materials Science and Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, USA

²Department of Materials Science and Engineering, University of Wisconsin, Madison, Wisconsin 53706, USA

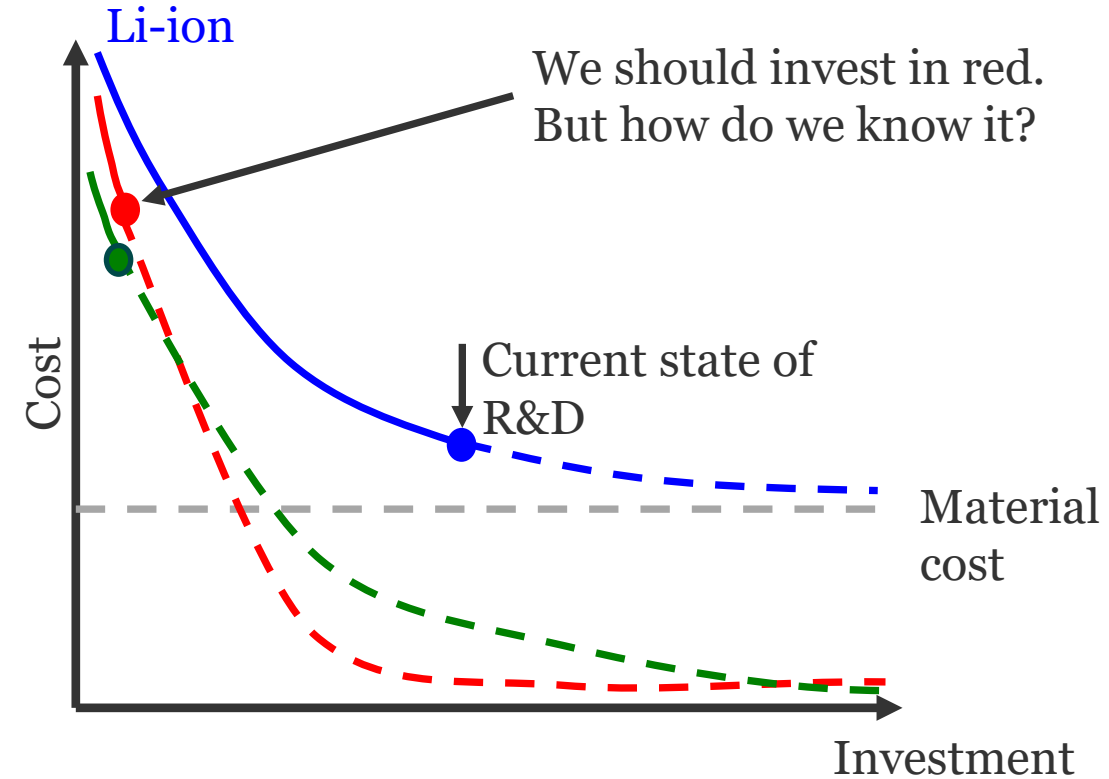
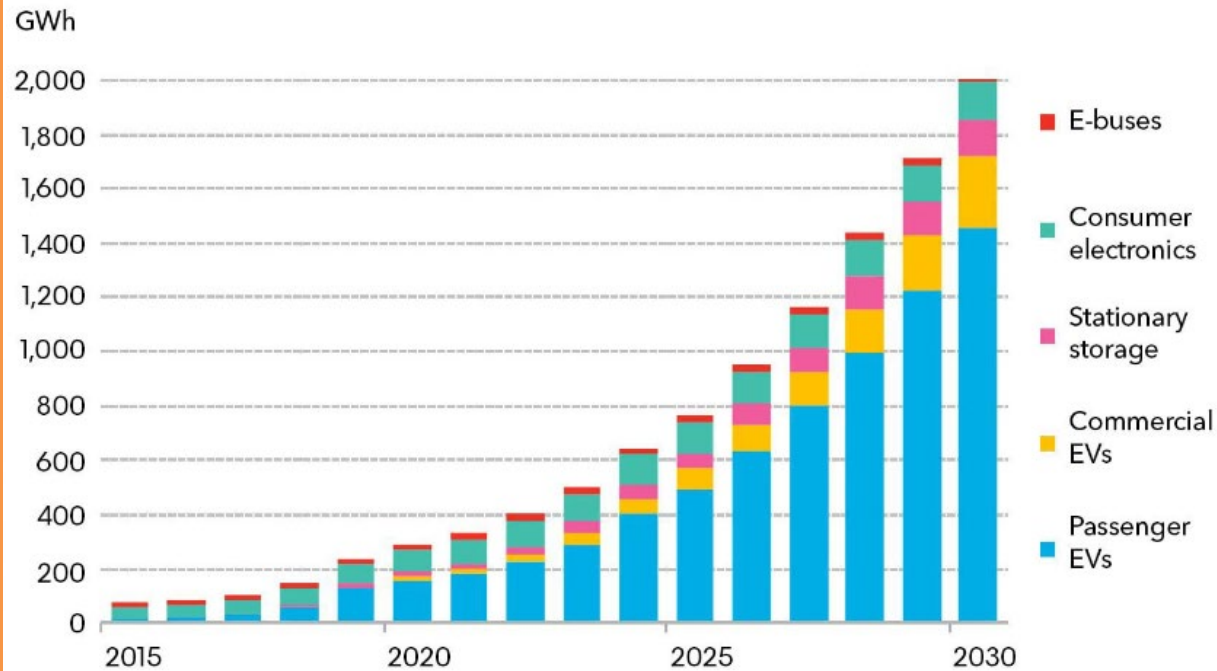
*e-mail: gceder@mit.edu

- **“Improve”**: Renewable energy, self-driving cars, transparent displays, memory technologies
- **“Discover”**: Room temperature superconductivity, high mechanical stress materials
- **“Engineer”**: Quantum computing, single-atom catalysts, biomolecules

Functionality, manufacturability, cost

Batteries: Li-ion and Beyond

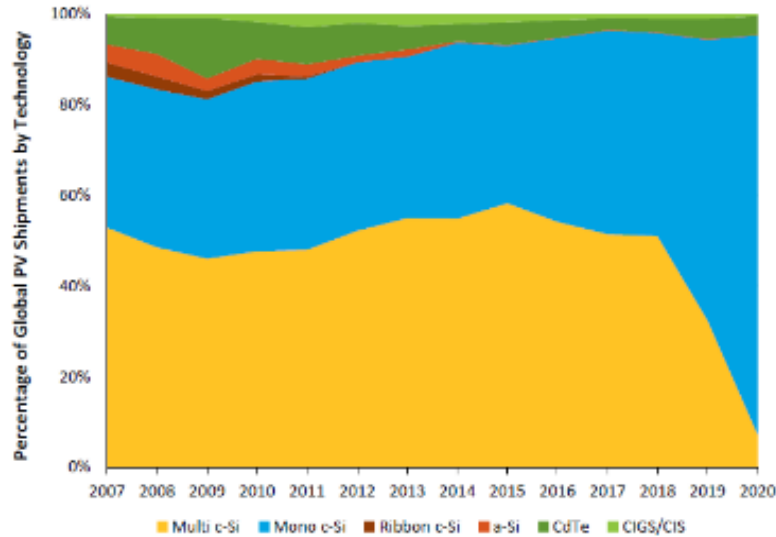
Annual lithium-ion battery demand



- Batteries are required element of energy transition (EVs, ESS, mobile devices)
- Currently Li-ion is the primary technology
- Optimization of Li-ion batteries takes years (even with same process on new Gigafactory)
- However, it is far from Goldilock zone for ESS or energy transport
- How can we optimize usage and safety for Li-ion batteries in EVs?
- How do we select beyond Li technologies for ESS?

Solar Energy: Will Silicon Ever Reign?

Global Annual PV Shipments by Technology*



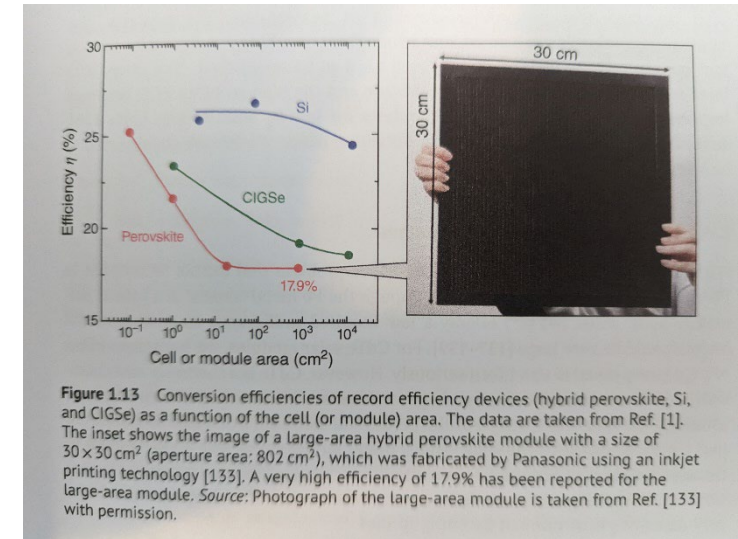
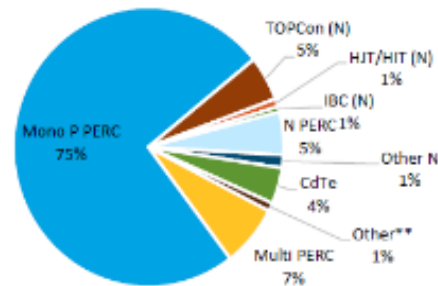
*Notes: Excludes inventory sales and outsourcing.

** Includes "Standard Multi c-Si", "Standard Mono c-Si", "a-Si", and "CIS/CIGS."

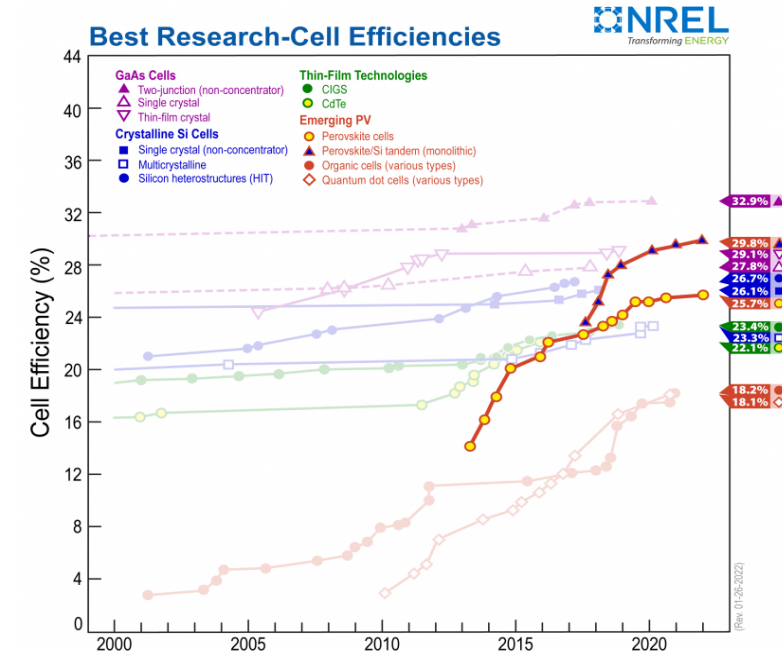
Source: 2004-2020: Paula Mints, "Photovoltaic Manufacturer Capacity, Shipments, Price & Revenues 2020/2021." SPV Market Research, Report SPV-Supply9, April 2021.

- In 2020, 88% of PV shipments were mono c-Si technology, compared to 35% in 2015 (when multi peaked at 58%).
- Mono P PERC was the dominant cell type in 2020, though n-type shipments grew 181%, y/y, to 13% of the market.

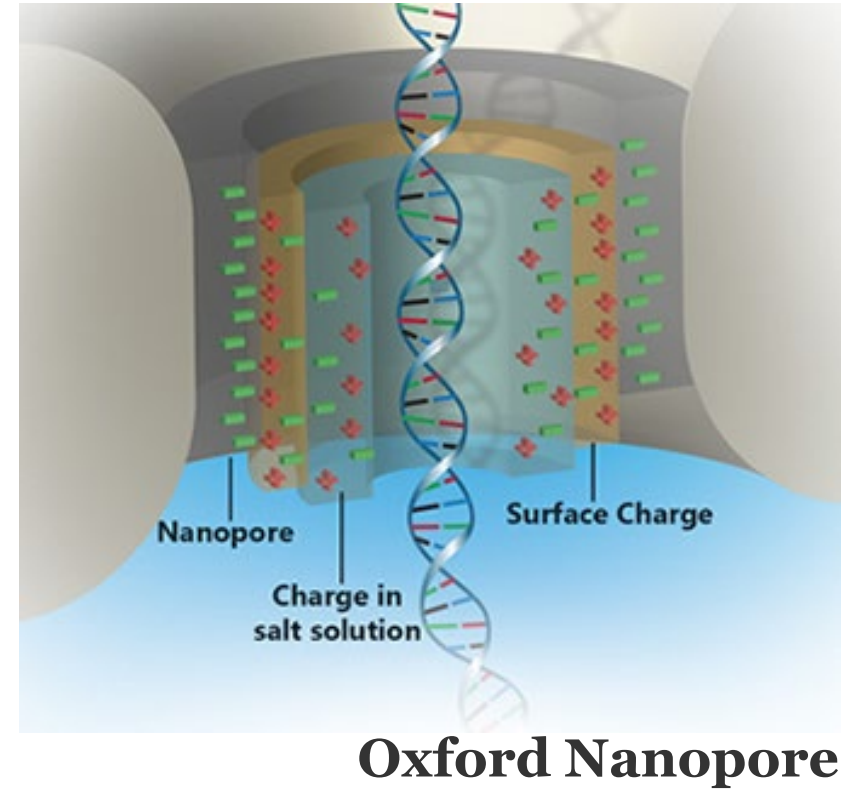
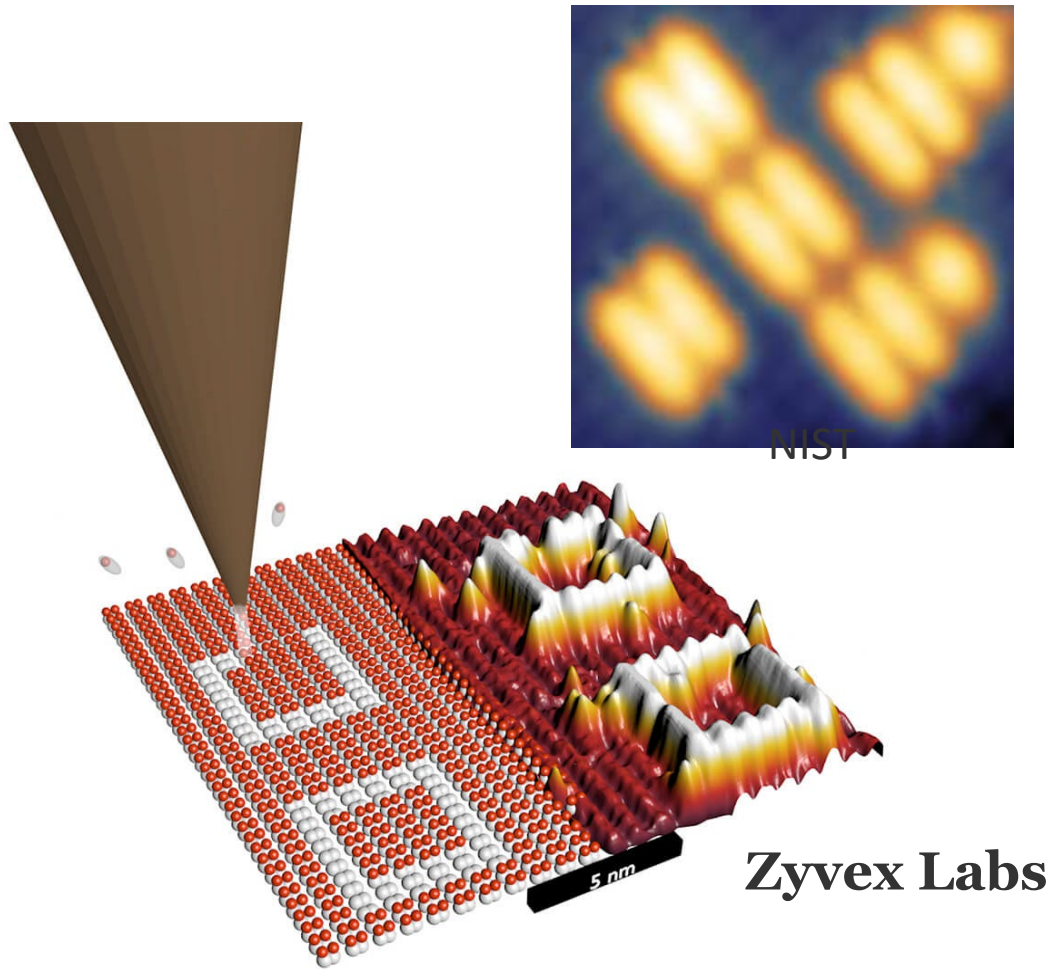
2020 Market Share by Cell Type



- Solar energy is the fastest growing energy sector
- Si is now reigning material – however, it is really not the optimal material for PV (heavy, expensive)!
- Hybrid perovskites can be used as ideal PV materials – if we can make them stable and scale manufacturing!



Quantum Computing and Single Molecule Bio



- Direct atomic fabrication: quantum communications and quantum computing, environmental sensing
- Single-molecule biological devices
- Success story 1: cryo-electron microscopy
- Success story 2: nanoelectron diffraction



Winning the Race

AMERICA'S AI ACTION PLAN

JULY 2025

"Today, a new frontier of scientific discovery lies before us, defined by transformative technologies such as artificial intelligence... Breakthroughs in these fields have the potential to reshape the global balance of power, spark entirely new industries, and revolutionize the way we live and work. As our global competitors race to exploit these technologies, it is a national security imperative for the United States to achieve and maintain unquestioned and unchallenged global technological dominance. To secure our future, we must harness the full power of American innovation."

Donald J. Trump
45th and 47th President of the United States



https://en.wikipedia.org/wiki/Michael_Kratsios



https://en.wikipedia.org/wiki/Lynne_Parker

Introduction

The United States is in a race to achieve global dominance in artificial intelligence (AI). Whoever has the largest AI ecosystem will set global AI standards and reap broad economic and military benefits. Just like we won the space race, it is imperative that the United States and its allies win this race. President Trump took decisive steps toward achieving this goal during his first days in office by signing Executive Order 14179, “Removing Barriers to American Leadership in Artificial Intelligence,” calling for America to retain dominance in this global race and directing the creation of an AI Action Plan.¹

Winning the AI race will usher in a new golden age of human flourishing, economic competitiveness, and national security for the American people. AI will enable Americans to discover new materials, synthesize new chemicals, manufacture new drugs, and develop new methods to harness energy—an industrial revolution. It will enable radically new forms of education, media, and communication—an information revolution. And it will enable altogether new intellectual achievements: unraveling ancient scrolls once thought unreadable, making breakthroughs in scientific and mathematical theory, and creating new kinds of digital and physical art—a *renaissance*.

An industrial revolution, an information revolution, and a renaissance—all at once. This is the potential that AI presents. The opportunity that stands before us is both inspiring and humbling. And it is ours to seize, or to lose.

 RELEASES

Trump Administration Announces More Than \$5 Billion for the Genesis Mission, a National Mission on AI for Science

The White House

July 22, 2026

 JOIN NOW

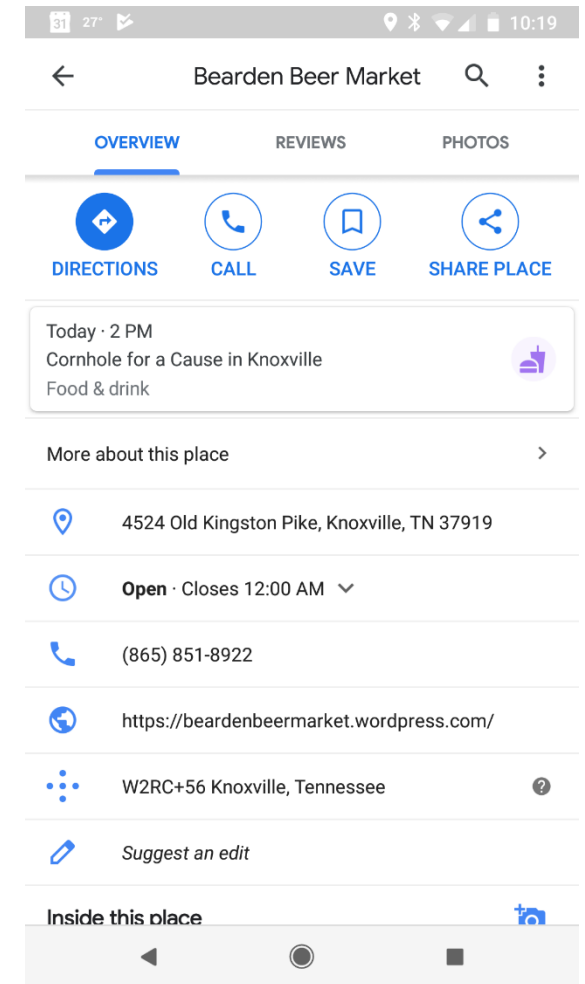
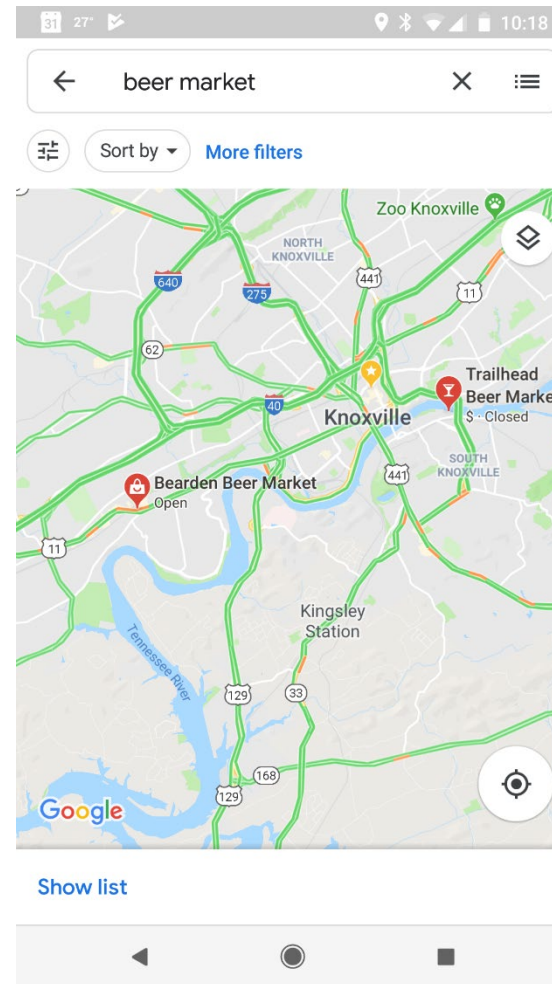
WASHINGTON, DC — Today, Assistant to the President and Director of the White House Office of Science and Technology Policy Michael Kratsios unveiled more than \$5 billion in Federal commitments expanding the [Genesis Mission](#) alongside new mission-driven [National Science and Technology Challenges](#). Tune in to the [Genesis Mission 2026 Summit](#) live on July



Click to listen

Modern day world

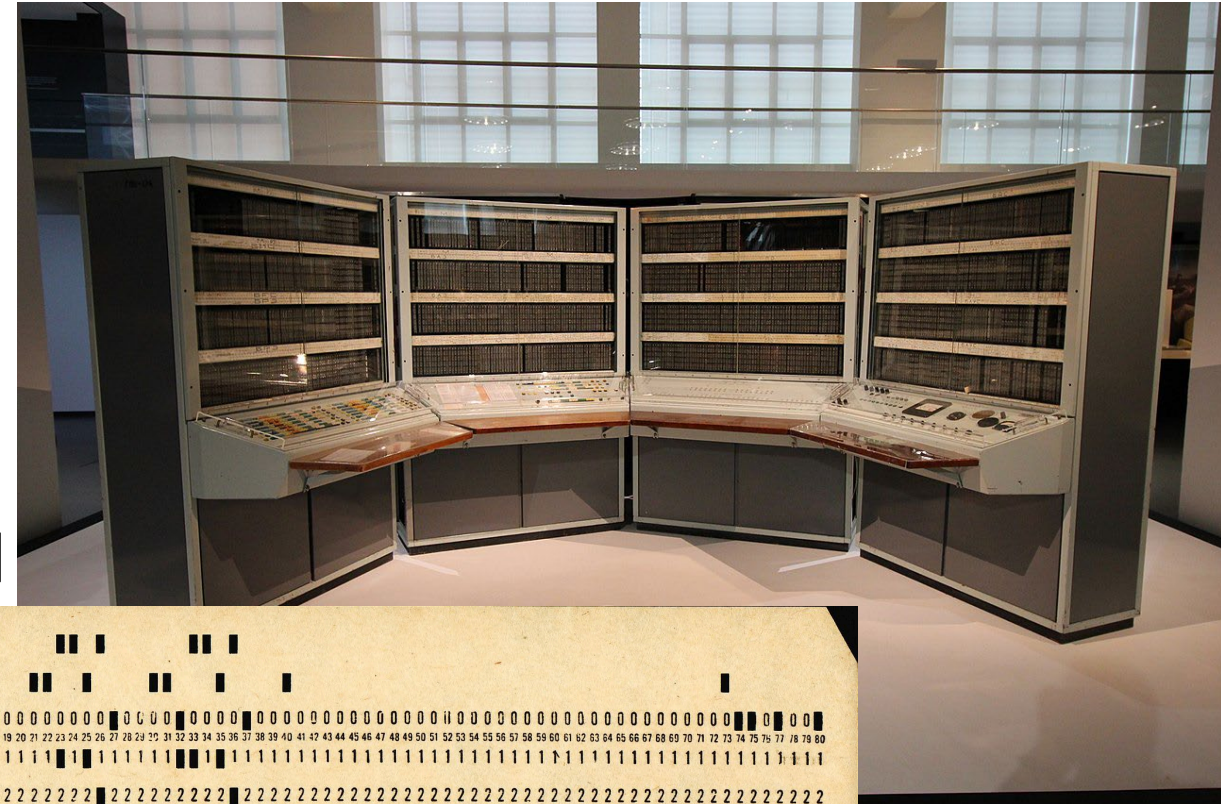
- Google
- Facebook
- Yelp
- Netflix
- Uber
- Lyft
- ...



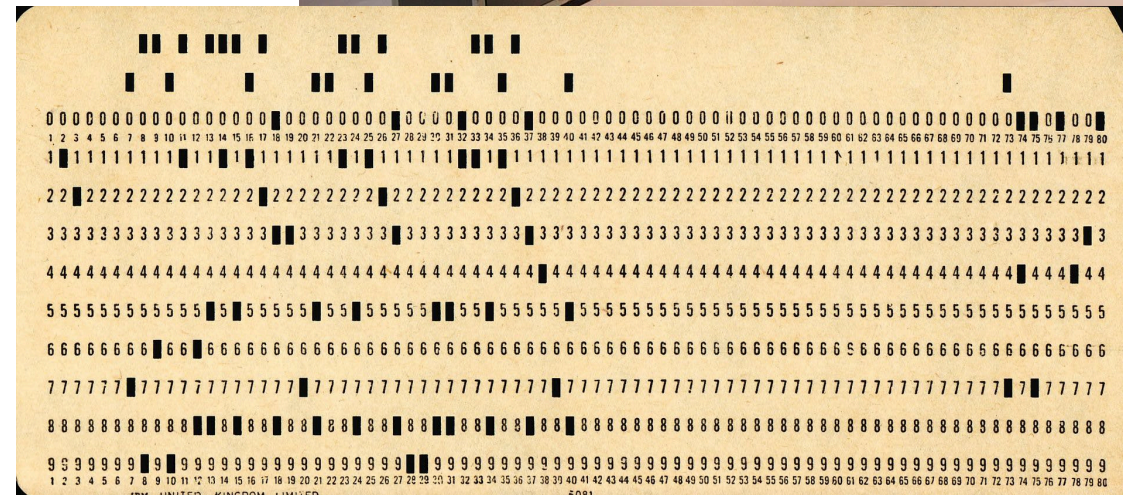
What was science like before 80ies?

BESM-6 Computer

- Computers existed only for specialized applications
- Many years of training before you can use one
- ... and even if you can, required much patience



Punch card



Altair(duino)



From Wikipedia

What if you want to know more?

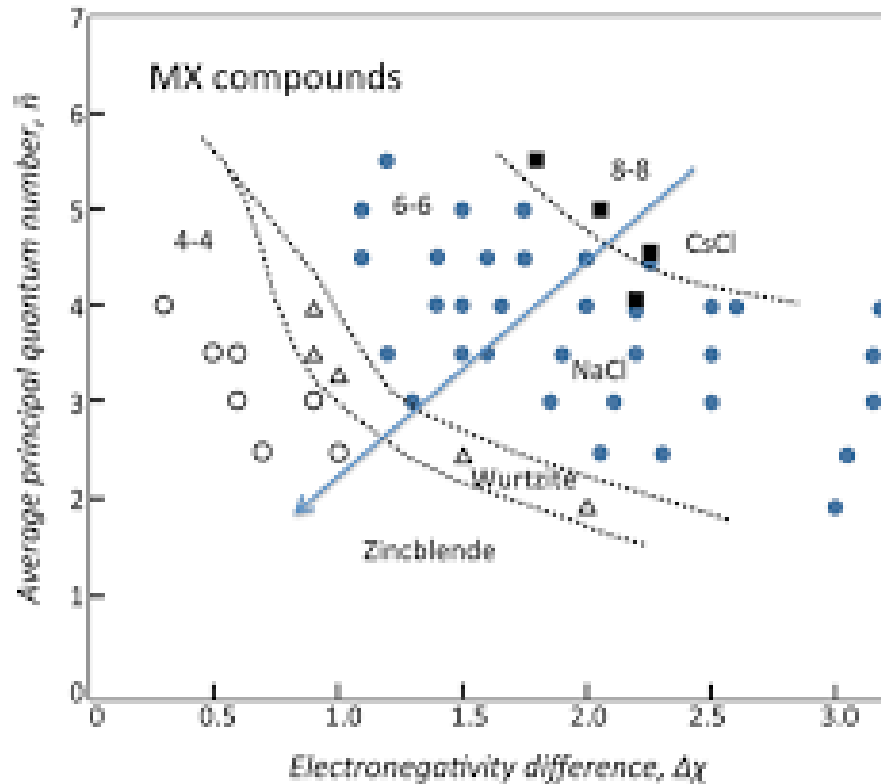


Libraries:

- Annual abstract books to find papers
- Dewey system to find books
- Collections: Landolt-Bornstein, etc

Was there machine learning for materials then?

Mooser-Pearson diagrams



- **Electronegativity:**

1	H 2.20																	He
2	Li 0.98	Be 1.57											B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne
3	Na 0.93	Mg 1.31											Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16	Ar
4	K 0.82	Ca 1.00	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 3.00
5	Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.6	Mo 2.16	Tc 1.9	Ru 2.2	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe 2.60
6	Cs 0.79	Ba 0.89	* Lu 1.27	* Hf 1.3	* Ta 1.5	* W 2.36	* Re 1.9	* Os 2.2	* Ir 2.20	* Pt 2.28	* Au 2.54	* Hg 2.00	* Tl 1.62	* Pb 1.87	* Bi 2.02	* Po 2.0	* At 2.2	* Rn 2.2
7	Fr >0.79	Ra 0.9	** Lr 1.3	** Rf [en 2]	** Db	** Sg	** Bh	** Hs	** Mt	** Ds	** Rg	** Cn	** Nh	** Fl	** Mc	** Lv	** Ts	** Og
			* La 1.1	* Ce 1.12	* Pr 1.13	* Nd 1.14	* Pm -	* Sm 1.17	* Eu -	* Gd 1.2	* Tb 1.1	* Dy 1.22	* Ho 1.23	* Er 1.24	* Tm 1.25	* Yb -		
			** Ac 1.1	** Th 1.3	** Pa 1.5	** U 1.38	** Np 1.36	** Pu 1.28	** Am 1.13	** Cm 1.28	** Bk 1.3	** Cf 1.3	** Es 1.3	** Fm 1.3	** Md 1.3	** No 1.3		

- **Nephelauxetic effect:**

$F^- < H_2O < NH_3 < en < [NCS - N]^- < Cl^- < [CN]^- < Br^- < N_3^- < I^-$

- **Solvent mixtures laws in organic chemistry**

Why does it matter? Physics and human heuristics is difficult to beat!

80ies

- First broadly available personal computers
- Can start programming (and get results) overnight
- First specialized scientific software
- What you have is what you bring



From Wikipedia

The 1980-2010 period:

Scientific information access:

- ISI and other data bases
- Electronic journals

General information

- Google and other search engines

Scientific software:

- Word
- Origin
- Digital micrograph

Programming languages

- MatLab
- Igor Pro

Instrument control (usually limited by manufacturer)

But relatively static....

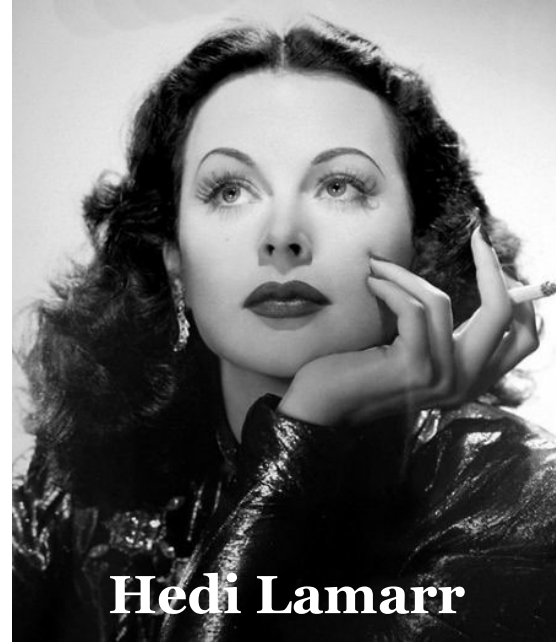
Could anyone have predicted it?



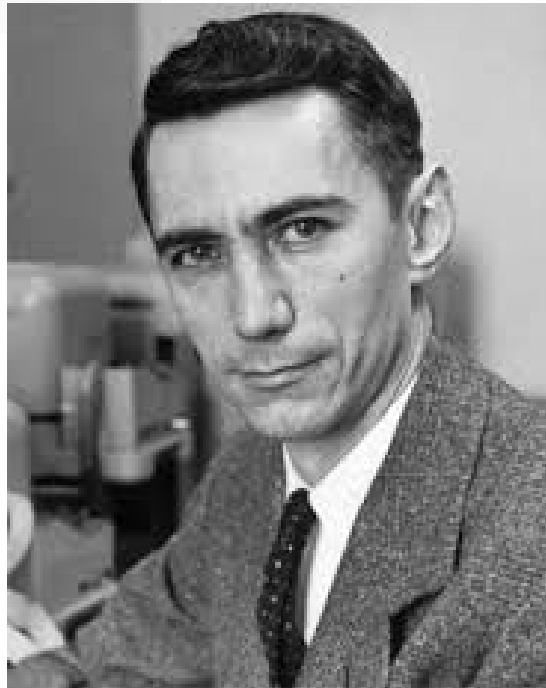
J. C. R. Licklider in 1965. A psycho-acoustician who saw computers as more than calculating machines, he was the first director of ARPA's Information Processing Techniques Office (IPTO). (Photo courtesy of the MIT Museum)



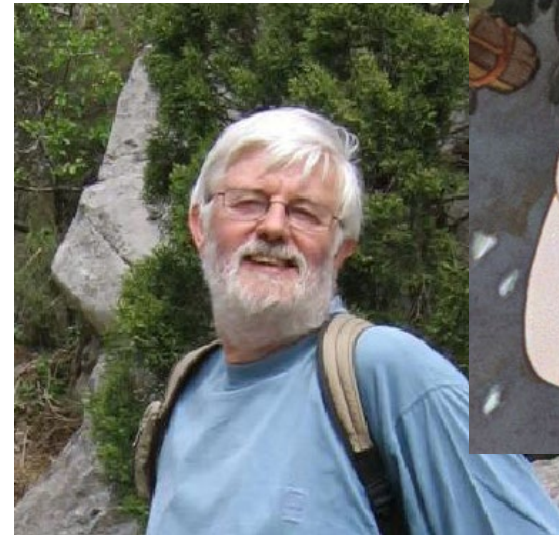
Ada Lovelace



Hedy Lamarr



Claude Shannon



Noel Bonnet



What is happening now?

New data technologies

- Searches
- Social networks
- Recommender engines
- Connection to real world
- Large Language models
- Agentic systems

New opportunities:

- 3D Printing
- IoT devices
- Laboratory robotics
- Open code
- Text analytics
- APIs, Orchestrators, MCPs

The inflection point (for theory): 2006

Predicting crystal structure by merging data mining with quantum mechanics

CHRISTOPHER C. FISCHER¹, KEVIN J. TIBBETTS¹, DANE MORGAN² AND GERBRAND CEDER^{1*}

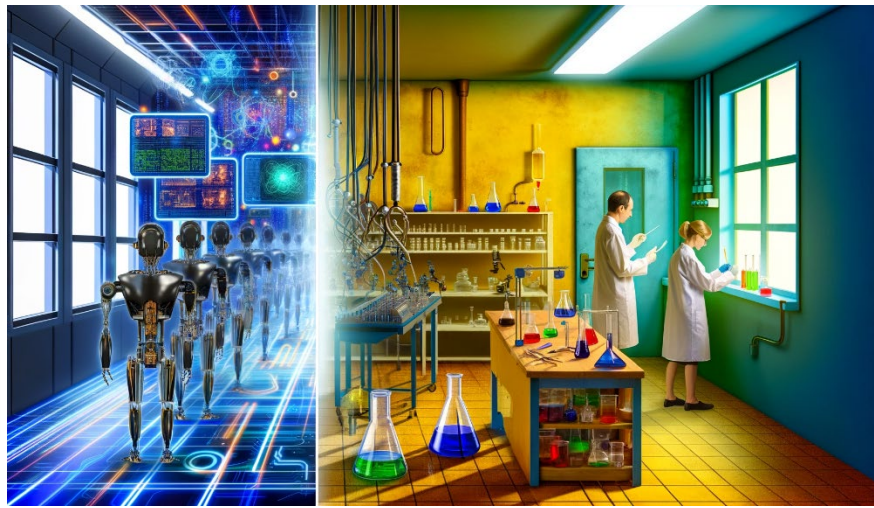
¹Department of Materials Science and Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, USA

²Department of Materials Science and Engineering, University of Wisconsin, Madison, Wisconsin 53706, USA

*e-mail: gceder@mit.edu

Publication by Gerd Ceder paper that is broadly seen as the inflection point launching Materials Genome Initiative in US and equivalent programs worldwide

Launch of AWS (Amazon Web Services) made cloud computing a reality – allowing businesses and scientists alike have access to computational resources without the need to build and maintain clusters



Machine learning in theory:

- Homogeneous workflows
- Known causal structure/lack of exogeneous factors
- Requires know-how, but relatively low entry barrier
- Easy to scale (given the funding)

Instrumentation

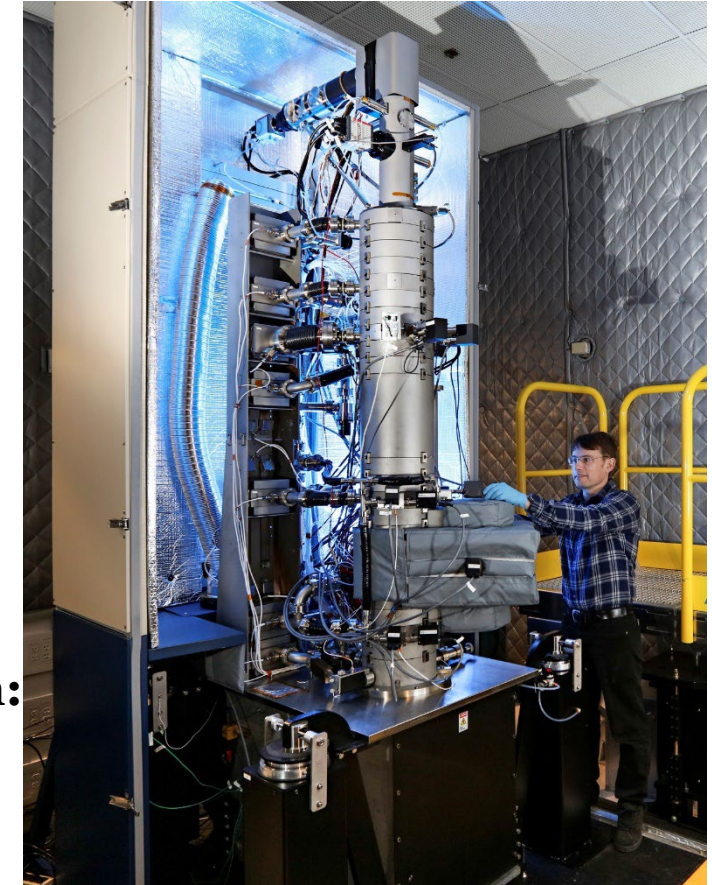


General synthesis/characterization:

- Multiple data generation tools
- Complex workflows
- 100s at each university in US

Surface science lab:

- Multiple data generation tools
- Complex controls and workflows
- 10s in each university

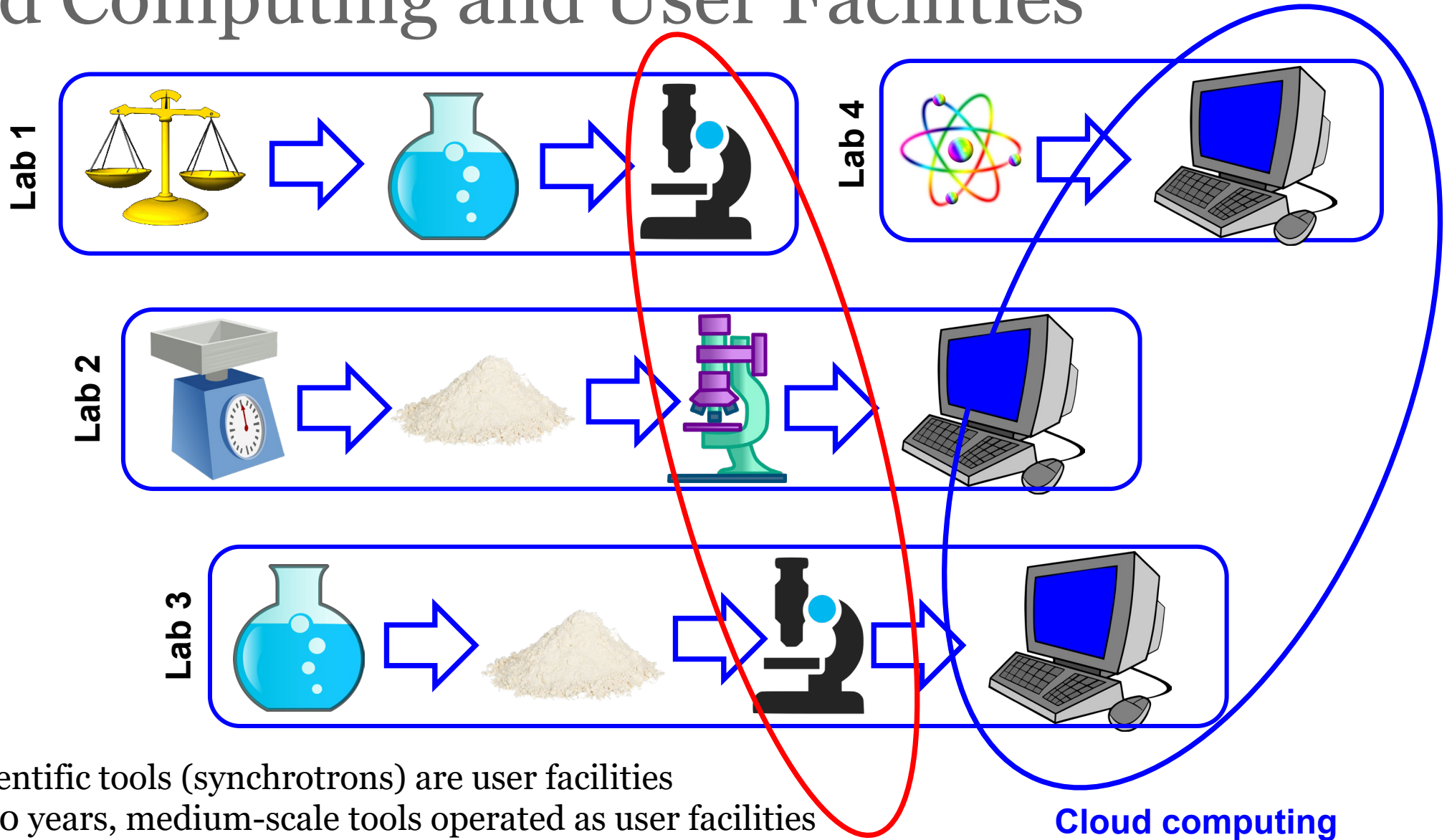


Electron microscope:

- >100k worldwide
- Can cost up to ~4-5\$M
- Can generate data at the ~10GB/s



Cloud Computing and User Facilities



- Big scientific tools (synchrotrons) are user facilities
- For ~20 years, medium-scale tools operated as user facilities
- Enterprise computing -> cloud computing
- Over last 5 years, cloud labs are emerging
- What about the workflows?

The inflection point (for experiment): ~2020

- **Before 2010:** A number of (usually) confidential efforts in industry
- **2010 - 2015:** Early adopters and visionaries (Cronin, Maryama, Kusne, etc)
- **2015 – 2020:** The time of engineers
- **2020 – now:** Automated experimentation becomes broadly available with very low cost entry barriers

age/chemrxiv/article-details/64f94d43dd1a73847f598050

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Review of Low-cost Self-driving Laboratories: The "Frugal Twin" Concept

08 September 2023, Version 1

This is not the most recent version. There is a [newer version](#) of this content available

Review

Stanley Lo, Sterling Baird, Joshua Schrier, Ben Blaiszik, Sergei Kalinin, Helen Tran, Taylor Sparks, Alán Aspuru-Guzik [Show author details](#)

This content is a preprint and has not undergone peer review at the time of posting.

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Abstract

This review proposes the concept of a "frugal twin," similar to a digital twin, but for physical experiments. Frugal twins range from simple toy examples to low-cost surrogates of high-cost research. For example, a color-mixing self-driving laboratory (SDL) is a low-cost version of a costly multi-step chemical discovery SDL. We need frugal twins because they provide hands-on experience, a test bed for software prototyping (e.g., optimization, data infrastructure), and a low barrier to entry for democratizing SDLs. However, there is room for improvement. The true value of frugal twins can be realized in three core areas. Firstly, hardware and software modularity, secondly, purpose-built design (human-inspired vs. hardware-centric vs. human-in-the-loop), and thirdly state-of-the-art (SOTA) software (e.g., multi-fidelity optimization). We also describe the ethical benefits and risks that come with

Version History

Nov 16, 2023 Version 2
Sep 08, 2023 Version 1

Metrics

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10.26434/chemrxiv-2023-6z9mo

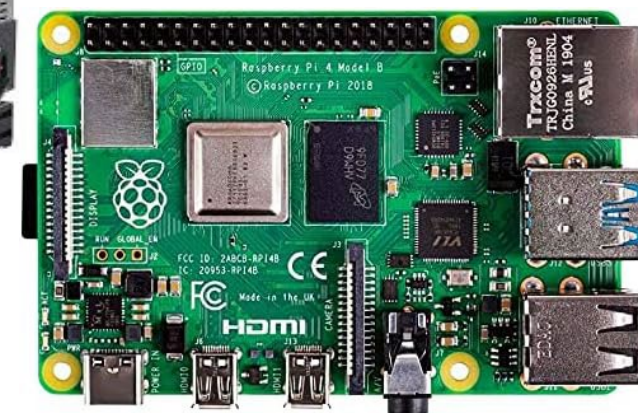
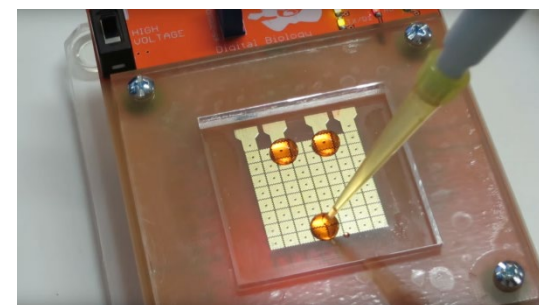
Funding

National Science Foundation
PHY-2226511

National Research Council Canada
509300

Division of Materials Research
1651668

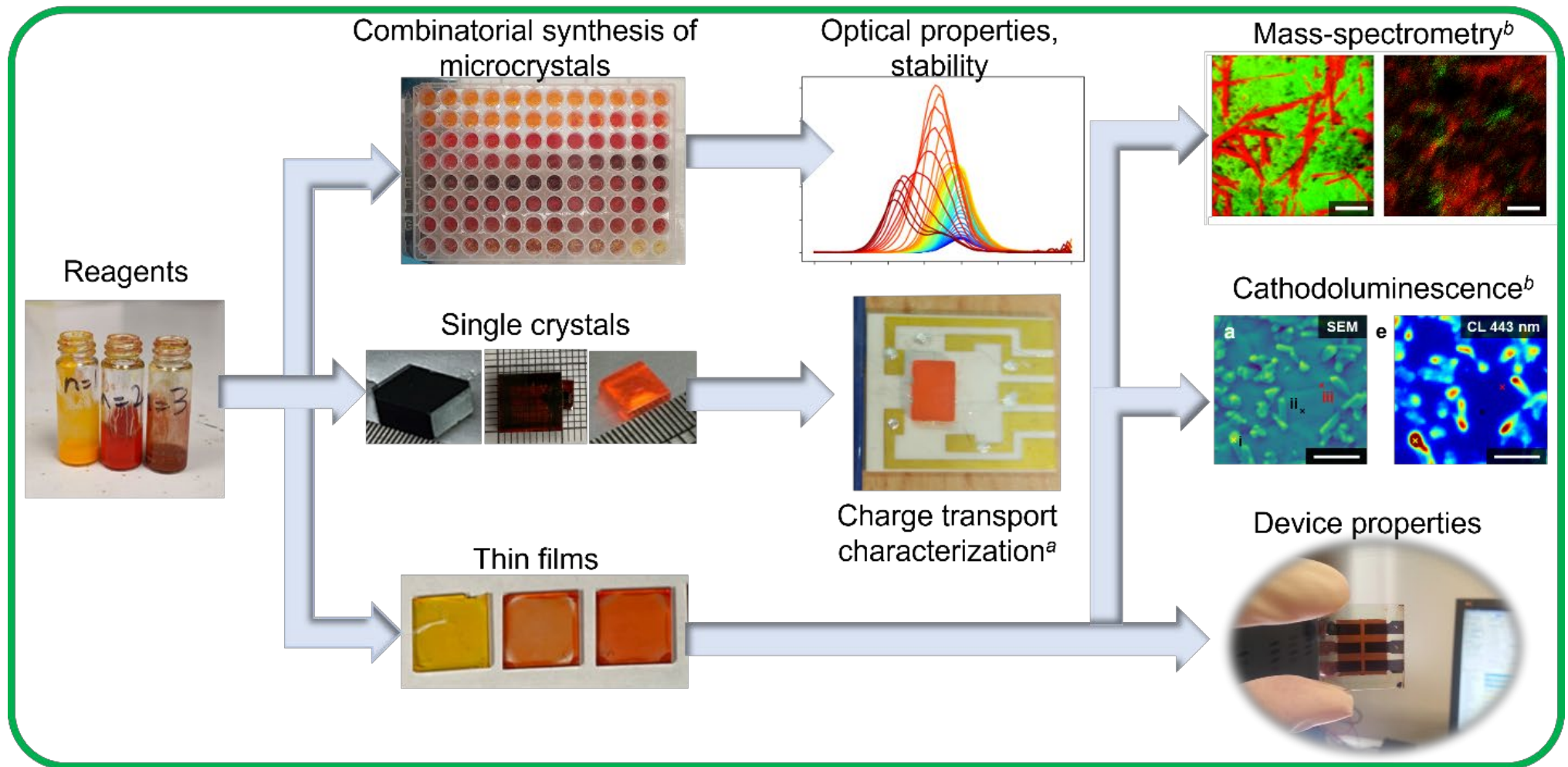
National Science Foundation
2334411



Why do we need machine learning?

To help us make decisions!

What is A Workflow?



- **Workflow:**
- Ideation, orchestration, implementation
- Domain specific language
- Dynamic planning: latencies and costs
- Reward and value functions

Designed in academia and adopted by industry

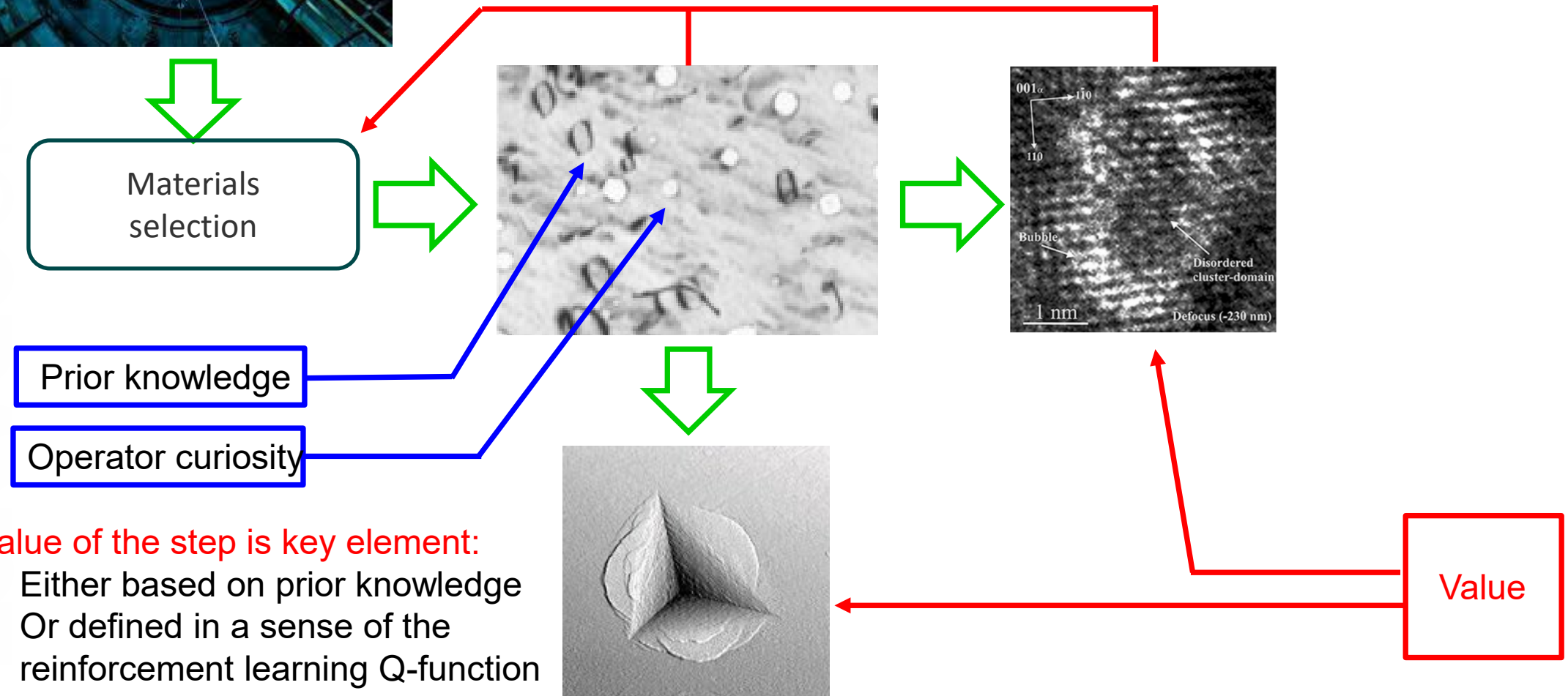
- Are they optimal?
- Can we design them better?
- Can they be changed dynamically?

Workflows for Nuclear Materials Design



Traditional experiment:

1. Always based on workflows
2. Ideated, orchestrated, and implemented by humans
3. The “gain of value” during the workflow implementation is uncertain



Value of the step is key element:

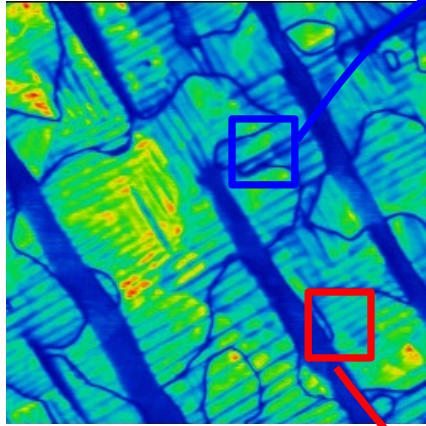
- Either based on prior knowledge
- Or defined in a sense of the reinforcement learning Q-function

Workflows in Scanning Probe Microscopy

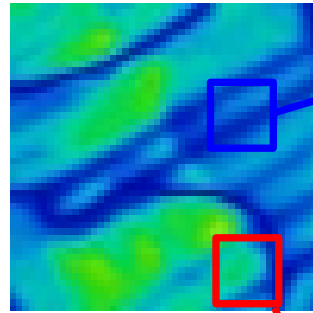
Workflow Plane

Workflow plane

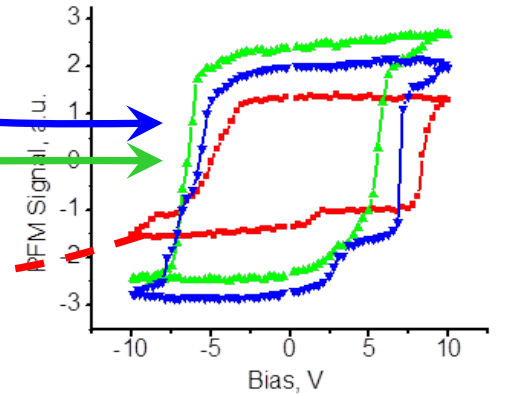
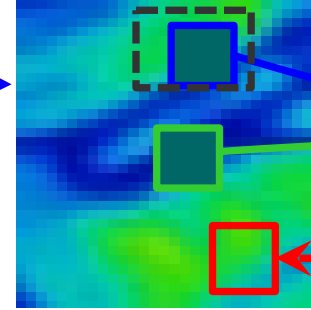
Overview scan



Zoom in



Zoom in

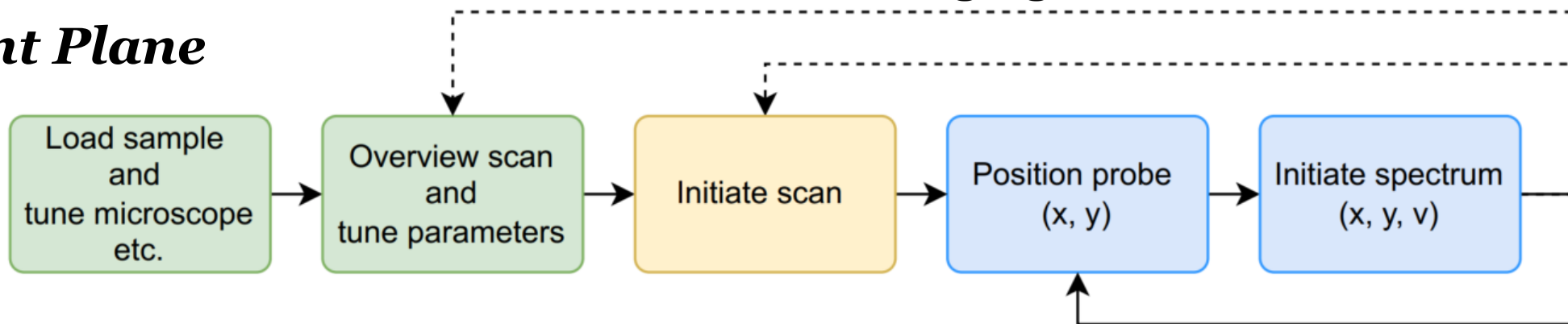


After acquisition analysis

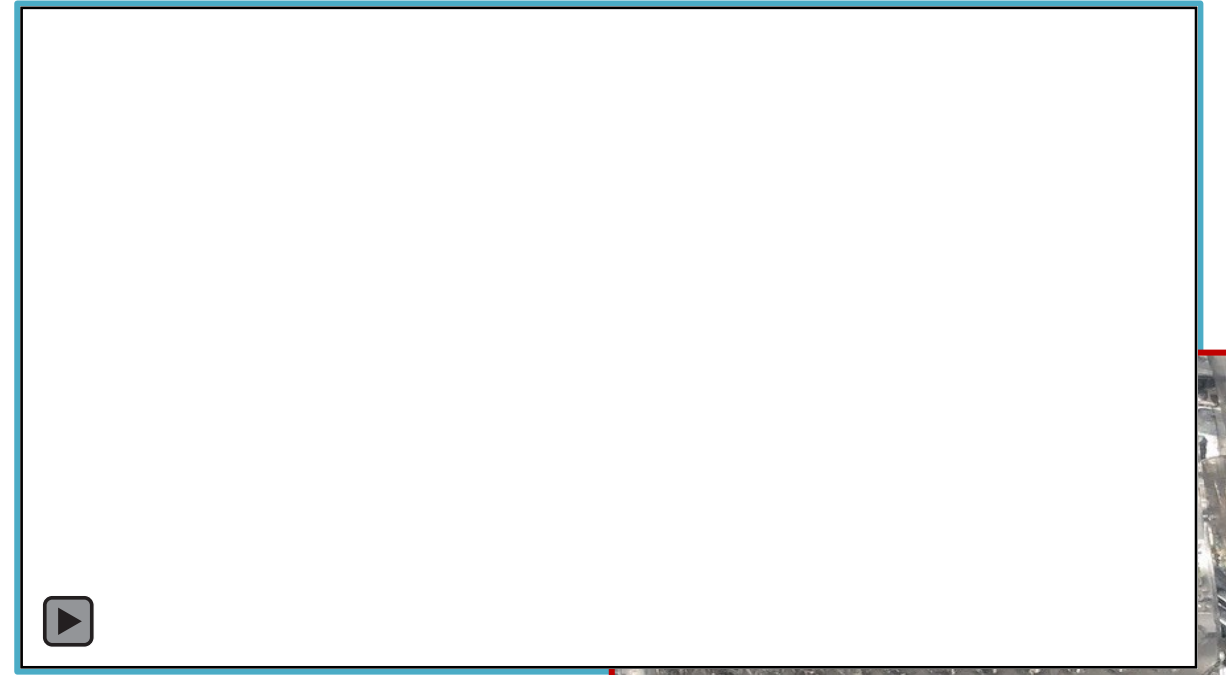
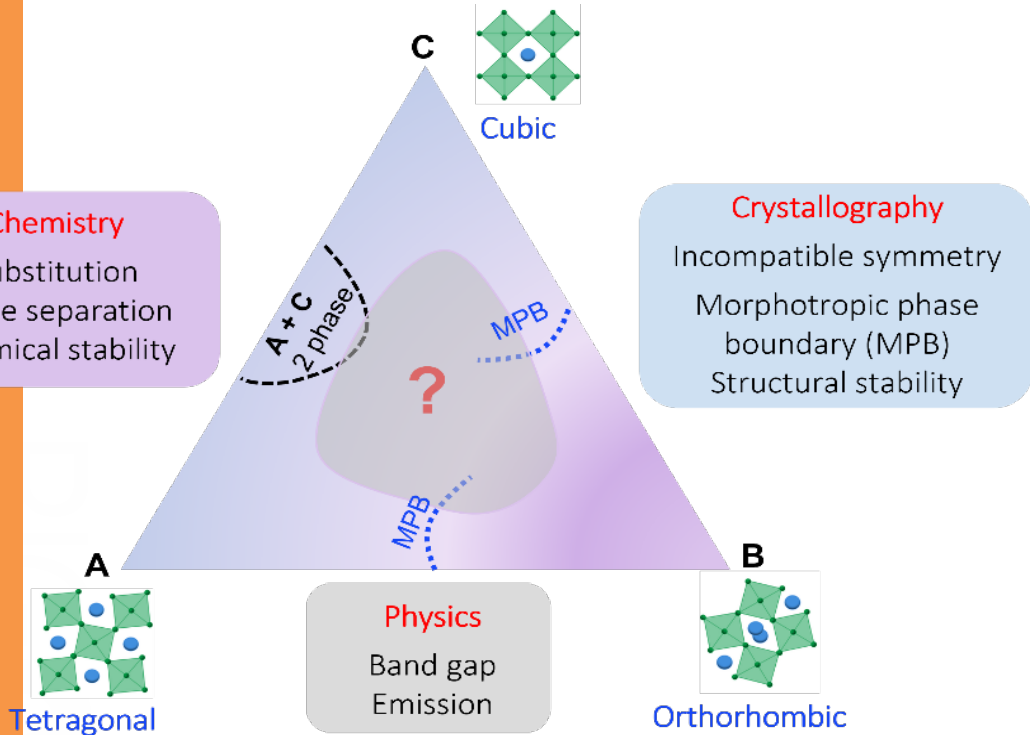
Instrument Plane

Instrument plane

Minimal instruction set control language

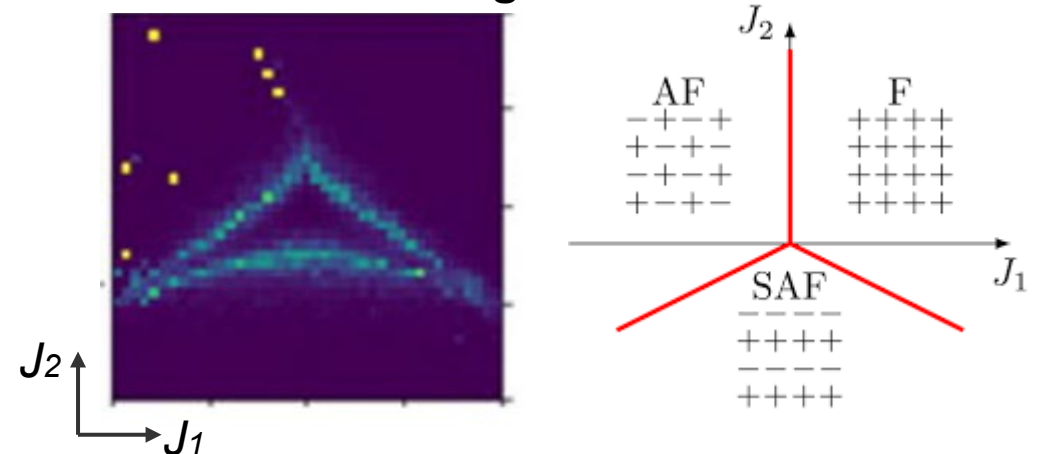


Why synthesis (or theory)?

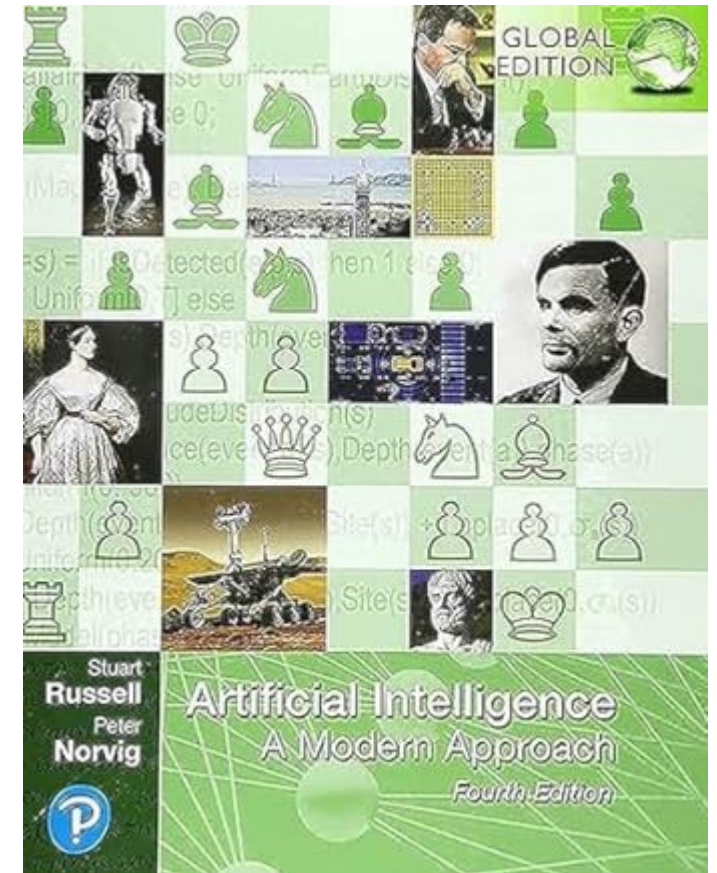


- Automated synthesis in its simplest form requires some way to navigate phase diagrams
- In more complex form, processing space.
- Ideally, incorporate physical knowledge
- Similar problem - theory

Ising model



Somewhat remarkably, almost all AI research until very recently has assumed that the performance measure can be exactly and correctly specified in the form of utility or reward function



In 2012-2014, large number of theory-trained workforce (DFT, MD, FEA) has moved into AI/ML fields

Many are returning to their original domain areas, building start-ups and corporate research centers. Many of these gain valuation very rapidly (In Silico, Schroedinger, etc)

The value created by AI/ML is orders of magnitude below investment (autonomous driving, radiology, cashier less stores)

Real-world industries and manufacturing are looking for work force with the combination of **domain** and **ML/AI** expertise

Course Information

Faculty Contact Information:

Instructor: Prof. Sergei V. Kalinin,
Office: 314 IAMM
E-mail: sergei2@utk.edu
Teaching Assistant: Kamyar Barakati, kbarakat@vols.utk.edu

Instructor Availability:

Please don't hesitate to email me with updates, questions, or concerns. I will typically respond within 24 hours during the week and 48 hours on the weekend (unless I travel). I will notify you if I will be out of town and if connection issues may delay a response.

Meeting Time: 10:20 am - 11:10 am MWF, Ferris Hall 511

The lectures and materials will be posted on Canvas and at GitHub:

https://github.com/SergeiVKalinin/MSE_Fall_2026

Office Hours:

Friday 1:30 - 3:00 PM are open for 1:1 meetings to discuss any course related item. Please set up time via email.

Prerequisites

To be successful in this course you will need a general background in materials science. Python or similar programming experience, while not essential, will be extremely useful. Students without any prior programming experience should expect to spend extra time outside of class learning basic skills.

This and that

Learning Environment:

The class will be delivered as in-person lectures. The Jupyter notebooks, code libraries, and videos provided. Weekly programming exercises will be assigned via Google Colabs and those students wishing to interact with the instructor in person should attend office hours.

Use of ChatGPT:

Strongly encouraged both for programming and written assignments. However, the students have to be aware of the limitations of the generative models.

Grading & Policies:

- Homework assignments 40%
- Mid-terms (2) 30%
- Final Project & Presentation 30%

Reference Materials

I will provide copies of lecture notes, presentations, and Colabs on GitHub and Canvas. There is no specific textbook for the course and we will take material from a variety of sources including:

- Andrew Bird et al, *Python Workshop – Second Edition*,
<https://subscription.packtpub.com/book/programming/9781804610619/1>
- Sebastian Raschka, *Machine Learning with PyTorch and Scikit-Learn*,
<https://subscription.packtpub.com/book/data/9781801819312/1>
- Rowel Atienza, *Advanced Deep Learning with TensorFlow 2 and Keras - Second Edition*,
<https://www.packtpub.com/product/advanced-deep-learning-with-tensorflow-2-and-keras-second-edition/9781838821654>
- (Optional) Alaa Khamis, *Optimization Algorithms: AI Techniques for Design, Planning, and Control Problems*, <https://www.manning.com/books/optimization-algorithms>
- (Optional) Peter Norvig, *Artificial Intelligence: A Modern Approach*, Global Edition,
<https://www.amazon.com/Artificial-Intelligence-Modern-Approach-Global/dp/1292401133>

Homework 1:

- Create new Colab, <https://colab.google/>
- Explore NotebookLM, <https://notebooklm.google/>
- Chapter 1-4, Python Workshop.